CAPSTONE PROJECT 1 REPORT

UPI Transactions & Fraud Risk Analysis

Domain : Data Science and Analytics With Generative AI

By: Swapna Gundeboina

Index

1. Introduction
2. Problem Statement
3. Objective
4. Tools & Technologies
5. Methodology
6. Data Description
7. Dashboard Design
8. Key Insights
9. Challenges and Limitations
10. Real-Life Applications
11. Conclusion
12. Future Scope
13. References

1. Introduction

Unified Payments Interface (UPI) has become one of the most widely used ways to move money in India. It allows instant bank-to-bank transfers through mobile apps such as Google Pay, PhonePe, Paytm and BHIM. As the number of payments grows, so does the risk of fraud. Banks, payment apps and risk teams need a clear way to see how much fraud is happening, where it is happening and which kinds of payments are most affected.

This project analyses a dataset of 25,000 UPI transactions from the year 2025 and presents the results in an interactive Power BI dashboard. The dashboard helps decision-makers understand transaction volume, transaction value and fraud patterns across months, cities, payment apps and transaction types.

2. Problem Statement

Digital payment platforms process a very large number of transactions every day, and a share of them turn out to be fraudulent. Without a visual and interactive view of the data, it is difficult for risk teams to answer basic questions: What is the overall fraud rate? Which payment apps, cities or transaction types carry more fraud? Do fraudulent payments differ from legitimate ones in value? The lack of a simple, filterable summary makes it harder to monitor fraud and to decide where controls are needed.

3. Objective

The primary goals of this project are to:

- Analyse UPI transaction data to understand overall volume, value and fraud level.
- Visualise key indicators such as total transactions, fraud transactions, fraud rate and average transaction amount.
- Compare legitimate and fraudulent transactions across cities, payment apps, months and transaction types.
- Give stakeholders an interactive dashboard with filters so they can monitor fraud and make data-driven decisions.

4. Tools & Technologies

Tool	Purpose
Power BI Desktop	Building the interactive three-page dashboard with KPI cards, charts and slicers
UPI transaction dataset	Source data containing 25,000 transaction records for 2025
Slicers and filters	Slicing the report by city, month, payment app, merchant category, transaction type and fraud status

The report uses a range of visuals: line chart, donut charts, bar and column charts, funnel chart, scatter plot, map, gauge and a detail table.

5. Methodology

1. **Data Loading:** The UPI transaction dataset was loaded into Power BI Desktop.
2. **Data Preparation:** Fields such as year, month, city, payment app, transaction type, amount and fraud status were reviewed so that they could be used for grouping and filtering.
3. **KPI Calculation:** Key indicators were calculated: total transactions, fraud transactions, legitimate transactions, fraud rate, total transaction amount and average transaction amount.

4. **Visual Analysis:** Charts were built to compare volume and value across months, cities, payment apps and transaction types, with fraud and legitimate transactions shown separately.
5. **Dashboard Development:** The visuals were arranged into three pages (Overview, Risk Signals and Transactions) and connected with common slicers so that every page responds to the same filters.

Fraud rate is calculated as:

Fraud Rate = Fraud Transactions ÷ Total Transactions × 100

For this dataset: 4.24K ÷ 25K × 100 = 16.96%.

6. Data Description

Source: UPI transaction dataset (2025)

Total Records: about 25,000 transactions

Key Fields:

Field	Description
Transaction_ID	Unique identifier of each transaction
Year and Month	Period in which the transaction took place (2025, January to December)
Sender_City	City of the sender: Mumbai, Delhi, Bengaluru, Hyderabad, Chennai, Kolkata, Pune or Other
Payment_App	App used: Google Pay, PhonePe, Paytm, BHIM or Other UPI App
Transaction_Type	P2P, P2M, Bill Payment or Recharge
Merchant_Category	Category of the merchant (available as a filter in the report)
Total Transaction Amount	Value of the transaction in rupees
Fraud_Status	Yes or No label showing whether the transaction was fraudulent
Unusual_Device_Flag	Flag (0 or 1) showing whether an unusual device was involved; the report contains further flag columns

Headline figures:

Measure	Value
Total transactions	25K
Fraud transactions	4.24K
Legitimate transactions	20.76K
Fraud rate	16.96%

Measure	Value
Total transaction amount	₹21.5M
Average transaction amount	₹861.49

7. Dashboard Design

The dashboard is built as three pages, all sharing the same filters on the left: Sender City, Year and Month, Payment App, Merchant Category, Transaction Type and Fraud Status. A row of six KPI cards is shown at the top of every page.

Sections:

1. **KPI Cards:** Total transactions, fraud transactions, legitimate transactions, fraud rate, total amount and average amount.
2. **Fraud vs Legitimate Donut:** Share of safe and fraudulent transactions.
3. **Trend Chart:** Number of transactions per month.
4. **Breakdown Charts:** Transactions by city, by payment app and amount by transaction type.
5. **Risk Visuals:** Fraud amount by type, risk distribution, app-wise fraud share, app risk ranking and city-wise fraud amount.
6. **Detail Page:** Record-level table, fraud map by city and fraud-rate gauge.
7. **Filters:** Slicing by city, month, app, merchant category, transaction type and fraud status.

Dashboard 1: Overview

The Overview page gives the big picture. It shows the six KPI cards, the monthly transaction trend (a line chart), the fraud versus legitimate split (a donut chart), transactions by city (a bar chart), payment app-wise transactions (a column chart) and the transaction amount by transaction type (a stacked bar chart).

Dashboard 2: Risk Signals

The Risk Signals page focuses only on fraud. It contains fraud amount by transaction type (a funnel chart), the transaction risk distribution (a column chart), payment app fraud analysis (a donut chart), the payment app risk ranking (a scatter plot of app volume against fraud rate) and the city-wise fraud amount (a column chart).

Dashboard 3: Transactions

The Transactions page is a detail view. It has a table of individual transactions (ID, year, month, city, app, type, amount, fraud status and device flag), a map of fraud transactions by city and a gauge that compares the 16.96% fraud rate with a 5% marker.

8. Key Insights

1. **High fraud rate:** 16.96% of transactions are fraudulent, about 3.4 times the 5% reference marker on the dashboard gauge.
2. **Steady monthly volume:** The dataset holds about 2.1K transactions a month. July is the busiest month (2,175) and February the quietest (1,927), with November also low (1,966).
3. **Fraud is spread across cities:** Fraud value ranges from ₹1.04M in Delhi to ₹0.84M in Pune, so no single city stands out.

4. **Two apps carry most of the volume and fraud:** Google Pay (1,487 cases, 35.06%) and PhonePe (1,479 cases, 34.87%) together hold about 70% of fraud cases, in line with their share of all transactions.
5. **Similar fraud rates across apps:** App fraud rates sit in a narrow band of roughly 16% to 18%. BHIM appears highest but has the fewest transactions.
6. **P2P carries the most fraud value:** Fraud amount is ₹3.19M for P2P, ₹2.55M for P2M, ₹1.21M for Bill Payment and ₹0.59M for Recharge.
7. **Fraud tickets are larger:** Fraud makes up about 35% of the total value but only 17% of transactions. Dividing by transaction counts gives roughly ₹1,780 per fraudulent transaction against ₹675 per legitimate one (derived from dashboard totals).

9. Challenges and Limitations

- Legitimate transactions (83%) far outnumber fraudulent ones (17%), so charts must be read with this imbalance in mind.
- Fraud and legitimate values differ widely in size, so counts and amounts can tell different stories.
- Small groups, such as BHIM with about 1.3K transactions, give less stable fraud rates than large ones.
- The dataset covers a single year (2025), so long-term trends cannot be judged.
- Fraud labels are taken as recorded in the data; the dashboard shows patterns and not the causes of fraud.
- Keeping the same filters working across all three pages required careful setup of the report.

10. Real-Life Applications

This dashboard can be used by:

- **Banks and Payment Apps:** To monitor fraud levels and refine security checks for different apps and transaction types.
- **Fraud and Risk Teams:** To spot patterns early and prioritise high-risk transactions for review.
- **Policy Makers and Regulators:** To support decisions on transaction limits and authentication rules.
- **Customer Awareness Teams:** To target alerts and education by app, city or payment type.
- **Data Analysts:** As a base to build fraud detection and risk scoring models.

11. Conclusion

This project shows how raw transaction data can be turned into a clear visual story about payment fraud. Using Power BI, the UPI dashboard gives an overview of volume and value, highlights fraud patterns across apps, cities and transaction types, and lets users drill down to individual records. The analysis shows a fraud rate of 16.96%, well above the 5% reference marker, spread across all segments, with fraudulent payments being larger on average than legitimate ones. The dashboard acts as a bridge between raw data and practical fraud-monitoring decisions.

12. Future Scope

1. **Predictive Modelling:** Use machine learning to predict which transactions are likely to be fraudulent.
2. **Risk Scoring:** Combine the device and other risk flags into a simple score for each transaction.
3. **Real-Time Data Integration:** Connect the dashboard to live data for continuous monitoring.
4. **Multi-Year Analysis:** Add more years of data to study long-term and seasonal trends.
5. **Merchant-Level Analysis:** Study fraud by merchant category in more detail.
6. **Alerts:** Set automatic alerts when the monthly fraud rate crosses a target such as 5%.

7. **Role-Based Dashboards:** Create separate views for executives, analysts and risk teams.
8. **Mobile Optimisation:** Publish the report for Power BI mobile for access at any time.

13. References

- UPI transaction dataset (2025) used in the Power BI report.
- Microsoft Power BI Desktop, used to build the dashboard.